

Predicting Cable Dynamics with Physical Attention Bias

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Abstract

Learned simulators for deformable linear objects (DLOs) such as cables have to predict the motion of cables they were not trained on and stay stable over long rollouts. Most of their error occurs where the cable touches itself or the floor. Attention over all pairs of cable segments can represent contact between parts of the cable that are far apart along its length, but attention has no notion of geometry. A cable has two pairwise distances, which agree only while it is straight: the arc-length distance along the cable, which governs elastic forces, and the Euclidean distance in space, which governs contact. We add a *physical attention bias*, an additive term on the attention logits with a learned rate, and ask which distance it should use. We compare no bias, each distance alone, and both distances on disjoint sets of heads, keeping the rest of the model and the training protocol fixed. A physical bias improves prediction on unseen cables. The gain is largest when attention is the only mechanism that connects distant segments: there, the arc-length bias reduces prediction error by 15% and more than halves the drift in segment length. The Euclidean bias alone stays close to unbiased attention, while assigning both distances across heads is best or near-best on every metric we report. Code and per-run records: <https://github.com/avihaig/dlogs>.

Keywords: graph transformers; attention bias; geometric priors; deformable linear objects

1. Introduction

A robot that manipulates a deformable linear object (DLO) such as a cable, rope, or suture needs a model of how it will move (Caporali et al., 2026; Yu et al., 2025a). Rod mechanics gives one (Bergou et al., 2008), but it is expensive to solve and needs material parameters that are hard to identify for a new cable. Learned dynamics models instead fit trajectories and are cheap to run online (Yang et al., 2021; Gu et al., 2025). They have to generalize in two ways. The cable met at deployment is not a training cable, and the model is trained on single steps but run in closed loop for hundreds of them, so its own errors become its next inputs.

Graph-based learned simulators update a graph by message passing, and the edge set decides which nodes interact. GNS rebuilds edges from a spatial radius at every step

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(Sanchez-Gonzalez et al., 2020), and MeshGraphNets adds world edges between nodes that are close in space but not connected in the mesh (Pfaff et al., 2021). For a DLO this choice matters. A cable is a one-dimensional chain in $\mathbb{R}^3$ and so has two distances: the *intrinsic* arc-length distance along the chain, which governs elastic forces, and the *extrinsic* Euclidean distance, which governs self-contact between segments that are close in space but far apart along the chain. The two agree while the cable is straight and diverge once it folds, and folding is where rollouts accumulate most of their error.

Hybrid graph transformers move this choice from the edge set into attention. A GraphGPS layer runs local message passing in parallel with dense attention over all node pairs (Rampásek et al., 2022), so either distance can enter the attention branch as a pairwise term without changing the graph. Graph transformers already add structure this way, through additive logit biases or attention masks, but mostly on static benchmarks and usually combining several signals (Ying et al., 2021; Luo et al., 2023; Liu et al., 2024; Yun et al., 2026). The closest hybrid model for DLOs aggregates local interactions before a Transformer encoder, for quasi-static 2D shape control (Yu et al., 2025b). Concurrent work on a cable suspended from a drone scales all attention logits by one state-dependent scalar computed from learned Lagrangian matrices (Pierallini et al., 2026); this changes how sharp attention is but adds no pairwise structure. So it is still open which distance should bias global attention during a rollout, and whether the answer changes when the local stream is removed.

This extended abstract is a first study of that question. We change only the pairwise distance and keep the backbone, optimizer, seeds, and evaluation windows fixed. We compare five variants with the same parameter budget, two of which are *dual-metric*: they put the two distances on disjoint subsets of heads. We evaluate each variant with the local stream on and off, and compare against chain-only, per-node MLP, and global LSTM baselines.

2. Method

Representation and target. The cable is $N=33$ vertices joined by $N-1$ bidirectional chain edges. Node features are a $C=70$ -frame velocity history, four material parameters, and the clipped floor distance; edge features are relative displacement, distance, and rest length. Absolute coordinates are never a feature, so the representation is invariant to translations parallel to the floor. The network predicts a per-node displacement; training is one-step with random-walk input noise (Sanchez-Gonzalez et al., 2020), evaluation autoregressive.

Hybrid block. Linear encoders lift node and edge features to width d , L identical layers follow, and a linear readout maps node embeddings to displacements. Each layer runs two streams on the *same* input, in parallel (Rampásek et al., 2022): a GatedGCN over the chain edges, which alone reads and updates edge embeddings, and dense multi-head attention over all N vertices, which sees node embeddings only. Each stream output is added to the layer input and normalized per node; the sum passes through a two-layer MLP with a second residual. Switching the GatedGCN off leaves global attention alone (*local off*); switching attention off leaves chain-only message passing (*chain only*).

Physical attention bias. We add one term to the attention logits, with a learned decay rate per head; the variants differ only in *which distance each head receives*. For head h the score and the attention weight are $s_{ij}^h = q_i^h \cdot k_j^h / \sqrt{d_k} - b_h(i, j)$, so after the softmax the

bias acts multiplicatively, $\alpha_{ij}^h \propto \exp(q_i^h \cdot k_j^h / \sqrt{d_k}) e^{-b_h(i,j)}$: $b_h=0$ recovers standard attention and a positive bias attenuates the contribution of vertex j to vertex i . Both metrics enter normalized to $[0, 1]$, $\tilde{C}_{ij} = |i - j| / (N - 1)$ and $\tilde{D}_{ij} = \|p_i - p_j\| / L$ with L the rest length, so their rates are comparable at initialization. The five variants set $b_h(i, j)$ to: *Unbiased*, 0 on every head; *Euclidean*, $\gamma_h \tilde{D}_{ij}$ on every head; *Chain*, $\beta_h \tilde{C}_{ij}$ on every head; *Mixed*, $\lfloor H/3 \rfloor$ heads Euclidean, $\lfloor H/3 \rfloor$ chain, the rest unbiased; and *Euclidean+Chain only*, half the heads Euclidean and half chain, none unbiased. The rates $\gamma_h, \beta_h = \text{softplus}(\theta_h) > 0$ are one learned scalar per biased head, initialized to one and shared across layers, so a prior can be softened or sharpened but not reversed. During rollout $\tilde{C}$ is fixed by the chain while $\tilde{D}$ is recomputed from the model’s *own* predicted positions, so the extrinsic bias sees the model’s errors while the intrinsic bias cannot drift. Mixed and Euclidean+Chain only are the two *dual-metric* allocations, identical except in whether the remaining heads stay unbiased. No variant adds a third distance or mixes both in one head; all differ by at most H scalars.

3. Experimental protocol

Data. Cables are simulated in MuJoCo 3.9.0 (Todorov et al., 2012): two grippers drive the cable to a randomized pose and release it, and the fall onto the floor is recorded at 500 Hz until it settles. Training uses 46 cables spanning rest length 0.80–1.60 m, diameter 1.5–10 mm, and five Young’s moduli from 10^6 to 10^9 Pa, 100 episodes each; the 40 test cables are interpolated inside the same hull and never appear in training, so the model must generalize to unseen cable instances and to long rollouts (Appendix A).

Training, rollouts, metrics. Every model trains for 100k steps at batch 512 with Adam at $d=128$, $H=8$, $L=4$ (891k parameters); the last checkpoint is reported and every hyperparameter was chosen on validation windows, never on test cables (Appendix B). Each model is rolled out for 400 predicted frames from the same 400 test windows. With $\hat{p}_i(t)$ and $p_i(t)$ the predicted and recorded positions at predicted frame t , the primary error is $\text{rel } \ell_2(t) = \frac{1}{NL} \sum_i \|\hat{p}_i(t) - p_i(t)\|_2$, averaged over the rollout. Two physical violations use the prediction alone: the link-length drift δ from the rest length $\ell_0 = L / (N - 1)$, and the self-intersection rate v , the fraction of frames in which two non-adjacent segments pass closer than the cable diameter. A predicted frame is *floor contact* when the recorded cable has a vertex at or below one diameter (Appendix C).

Comparison design. The matrix is the five variants at three seeds with the local stream on and off, plus three baselines at three seeds: 39 runs. The per-node MLP never mixes vertices; the global LSTM reads the flattened state as a C -step sequence and writes all displacements. A cell reports the mean of the three seed means $\pm$ the mean of the three window sds $\pm$ the sd of the seed means; we do not run significance tests.

4. Results

The largest effect in Table 1 comes from the architecture, not the bias. Every GPS row lowers mean $\text{rel } \ell_2$ relative to chain-only message passing, by 65–68% with the local stream on and 57–63% with it off, and the MLP and LSTM baselines are far behind as well. The two streams also fail in different ways. Removing the local stream raises $\text{rel } \ell_2$ by 7–23% but

	$\text{rel } \ell_2$ (frac. of L)	δ (frac. of ℓ_0)	v (frac. of frames)
<i>local stream on</i>			
Unbiased	$0.0325 \pm 0.0281 \pm 0.0058$	$0.0300 \pm 0.0159 \pm 0.0032$	$0.171 \pm 0.231 \pm 0.009$
Euclidean	$0.0320 \pm 0.0267 \pm 0.0042$	$0.0303 \pm 0.0135 \pm 0.0021$	$0.174 \pm 0.229 \pm 0.032$
Chain	$0.0316 \pm 0.0341 \pm 0.0057$	$0.0271 \pm 0.0124 \pm 0.0023$	$0.167 \pm 0.227 \pm 0.030$
Mixed	$0.0302 \pm 0.0251 \pm 0.0067$	$0.0267 \pm 0.0109 \pm 0.0030$	$0.168 \pm 0.229 \pm 0.012$
Euclidean+Chain only	$0.0297 \pm 0.0288 \pm 0.0035$	$0.0275 \pm 0.0129 \pm 0.0007$	$0.165 \pm 0.229 \pm 0.012$
<i>local stream off</i>			
Unbiased	$0.0399 \pm 0.0246 \pm 0.0002$	$0.1693 \pm 0.1891 \pm 0.0030$	$0.216 \pm 0.258 \pm 0.006$
Euclidean	$0.0382 \pm 0.0279 \pm 0.0007$	$0.1470 \pm 0.2232 \pm 0.0178$	$0.208 \pm 0.262 \pm 0.020$
Chain	$0.0339 \pm 0.0195 \pm 0.0008$	$0.0783 \pm 0.0382 \pm 0.0040$	$0.174 \pm 0.241 \pm 0.004$
Mixed	$0.0352 \pm 0.0211 \pm 0.0016$	$0.0779 \pm 0.0455 \pm 0.0036$	$0.172 \pm 0.242 \pm 0.016$
Euclidean+Chain only	$0.0343 \pm 0.0202 \pm 0.0002$	$0.0784 \pm 0.0409 \pm 0.0013$	$0.161 \pm 0.234 \pm 0.006$
<i>baselines</i>			
chain only	$0.0925 \pm 0.0838 \pm 0.0720$	$0.0812 \pm 0.0589 \pm 0.0322$	$0.115 \pm 0.191 \pm 0.024$
MLP	$0.0868 \pm 0.0785 \pm 0.0163$	$1.35 \pm 2.32 \pm 0.73$	$0.435 \pm 0.256 \pm 0.010$
LSTM	$0.1070 \pm 0.0911 \pm 0.0397$	$0.2819 \pm 0.3124 \pm 0.0509$	$0.082 \pm 0.150 \pm 0.031$

Table 1: Unseen-cable test, 400 windows of 400-frame rollouts; mean of seed means $\pm$ mean of window sds $\pm$ sd of seed means; bold: lowest mean per block.

multiplies drift by up to $5.6\times$, which suggests that dense attention provides the long-range coupling and local message passing holds the rest length.

Every physical bias beats unbiased attention on $\text{rel } \ell_2$ in both blocks, by more when attention is the only path between distant segments. With the local stream off, Chain reaches 0.0339 ± 0.0008 against 0.0399 ± 0.0002 , a 15% reduction with seed means that do not overlap, and it more than halves drift. With the local stream on, the largest gap shrinks to 9% and lies inside the window spread, so with three seeds it is only an ordering. The two distances are not interchangeable. With the local stream off, the extrinsic bias alone lowers $\text{rel } \ell_2$ by 4% and the intrinsic bias by 15%, and the intrinsic bias alone closes about two thirds of the drift gap that the whole GatedGCN stream closes.

Splitting the local-on rollouts by phase (Table 2) shows where the error is: $\text{rel } \ell_2$ during floor contact is about eight times that in free flight. No single distance is best everywhere. Chain is only third on local-on $\text{rel } \ell_2$ and worst in both phase columns, and Euclidean stays close to unbiased whenever the local stream is off. Euclidean+Chain only, by contrast, is lowest or within one seed standard deviation of the lowest in all six cells of Table 1, with Mixed close behind. We therefore suggest the dual-metric bias as a default.

5. Conclusion

A physical attention bias costs at most H learned scalars and closes part of the gap between the two streams: the intrinsic distance alone recovers about two thirds of the drift benefit of the whole local stream. The effect clearly exceeds seed noise only when attention is the sole path between distant segments. Our results tentatively favor the dual-metric bias, with both distances spread across heads and no head left unbiased. Caveats: the local-on ranking lies inside the window spread, test cables lie inside the training range, and a lowest mean does not by itself show that heads specialize.

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Appendix A. Data generation

Each cable is a 32-segment chain ($N=33$ vertices) under MuJoCo’s cable elasticity plugin with twist-to-bend ratio $G/E=0.38$, integrated with `implicitfast` at $\Delta t=4.2 \times 10^{-4}$ s; kinematics are recorded at 500 Hz (the actual rate is rounded to a whole number of simulator steps, so the time step is read from the recorded timestamps). The scene is a floor and two grippers welded to the cable ends. Each episode settles the cable, drives the grippers quasi-statically to a randomized pose (end separation between 0.10 m and a quarter of the rest length, a 0.5 m sampling box, a $\pi/4$ end-facing cone, a 25 s reach and a 5 s dwell), and releases both welds on the same tick; $t=0$ is that instant. Recording ends when the 95th percentile of vertex speed stays below 1 cm/s for 0.5 s, or after 5 s. Self-collision is permitted and common.

The training population is 46 cables in five material classes (Young’s modulus 10^6 , 10^7 , 10^8 , 5×10^8 , 10^9 Pa, each with its own density and joint damping), four rest lengths (0.80, 1.10, 1.40, 1.60 m), and diameters from 1.5 to 10 mm, each released from 100 randomized poses (two sets of 50, drawn with different seeds). The test population is 40 cables with interpolated lengths and diameters inside the same hull, 30 episodes each. Test windows start at a stride of the rollout length inside every episode; a stratified cap of 400 windows is taken round-robin across cables so that all models are scored on the same windows.

Appendix B. Architecture and training

Node features are $F = 3C + 5 = 215$ wide at $C=70$: the velocity history newest first, the four material parameters, and the floor distance clipped at 1 m. The material channels are encoded as $(\log_{10} x - c)/h$ with (c, h) equal to $(7, 3)$, $(-3, 2)$, $(-2.301, 1)$, and $(-1.3495, 2.6505)$ for Young’s modulus, joint damping, diameter, and linear density; velocity history, floor distance, edge features, and the displacement targets are standardized with statistics fitted on the training split before noise is added and frozen thereafter; positions are not scaled.

The block uses $d=128$, $H=8$, $L=4$, a two-layer ReLU MLP of hidden width $2d$, and post-residual LayerNorm over the feature axis of each node. The GatedGCN stream normalizes its aggregation by the gate sum. The loss is the mean squared error on the standardized one-step displacement with random-walk velocity noise of scale $\sigma=0.1$ m/s per axis at the newest frame. Adam runs 100k steps at batch 512 with an exponential decay from 10^{-3} to 10^{-4} , gradient clipping at 1.0, weight decay 0, and dropout 0; the reported checkpoint is the last one for every model and seed. The per-node MLP has width 2048 and three layers; the global LSTM has width 256 and three layers and receives centroid-relative positions divided by the rest length in addition to the node features. One run trains and is evaluated in under an hour on a single consumer GPU; the 39 runs total roughly 30 GPU-hours.

Appendix C. Metric definitions and the phase split

	floor contact rel ℓ_2	free flight rel ℓ_2
Unbiased	$0.0511 \pm 0.0536 \pm 0.0105$	$0.0062 \pm 0.0095 \pm 0.0001$
Euclidean	$0.0510 \pm 0.0615 \pm 0.0080$	$0.0062 \pm 0.0099 \pm 0.0005$
Chain	$0.0529 \pm 0.0871 \pm 0.0122$	$0.0064 \pm 0.0131 \pm 0.0014$
Mixed	$0.0484 \pm 0.0551 \pm 0.0135$	$0.0058 \pm 0.0097 \pm 0.0011$
Euclidean+Chain only	$0.0479 \pm 0.0643 \pm 0.0068$	$0.0061 \pm 0.0127 \pm 0.0010$

Table 2: Local-on rows of Table 1 split by phase of the recorded trajectory (400 and 345 eligible windows); bold: lowest mean per column. Every column ordering here sits inside the seed spread.

Let $\hat{p}_i(t)$ and $p_i(t)$ be the predicted and recorded positions of vertex i at predicted frame $t \in \{1, \dots, 400\}$ (frame 0 is the observed input state and is never scored) and L the sample’s rest length. With $e_i(t) = \|\hat{p}_i(t) - p_i(t)\|_2$ and $\ell_2(t) = \frac{1}{N} \sum_i e_i(t)$, $\text{rel } \ell_2(t) = \ell_2(t)/L$, and a rollout reports the mean of $\text{rel } \ell_2(t)$ over its predicted frames. Link-length drift and self-intersection use the prediction and the cable geometry only; recorded rollouts give the simulator’s own baseline on both. With $\ell_0 = L/(N-1)$ and $\hat{\ell}_k(t) = \|\hat{p}_{k+1}(t) - \hat{p}_k(t)\|_2$,

$$\delta(t) = \frac{1}{N-1} \sum_{k=1}^{N-1} \frac{|\hat{\ell}_k(t) - \ell_0|}{\ell_0}, \quad (1)$$

reported as the mean over the rollout. Let $d_{jk}(t)$ be the shortest distance between the centerlines of segments j and k and D the cable diameter; a pair with $|j-k| > 2$, beyond the two-edge bending stencil, violates when $d_{jk}(t) < D$, and the self-intersection rate is the mean over the rollout of $v(t) = \mathbf{1}[\min_{|j-k|>2} d_{jk}(t) < D]$. For the phase split a predicted frame is floor contact when $\min_i z_i(t) \leq D$ on the recorded trajectory after the input frame and free flight otherwise; each rollout contributes the mean of $\text{rel } \ell_2(t)$ over its selected frames, and a phase cell is the equal-weight mean over rollouts with at least one selected frame (400 floor contact and 345 free flight of the 400 windows). The per-seed window standard deviations and counts are taken from the same per-window scores.